6.3 SPEED OF LIGHT

Light travels with the highest speed in vacuum. The speed of light in vacuum is denoted as c and its value is, $c = 3 \times 10^8\ m\,s^{-1}$. The earliest attempt to determine the speed of light was made by a French scientist Hippolyte Fizeau (1819–1896). That paved way for the other scientists too to determine the speed of light.

6.3.1 Fizeau's method to determine speed of light

Apparatus: The apparatus used by Fizeau for determining speed of light in air medium is shown in Figure 6.13. The light from the source S was first allowed to fall on a partially silvered glass plate G kept at an angle of 45° to the incident light. The light then was allowed to pass through a rotating toothed-wheel with N teeth and N cuts of equal widths whose speed of rotation could be varied through an external mechanism (not shown in the Figure). The light passing through one cut in the wheel will get reflected by a mirror M kept at a long distance d, about 8 km from the toothed-wheel. If the toothed-wheel was not rotating, the light reflected back from the mirror would again pass through the same cut and reach the eyes of the observer who looks through the partially silvered glass plate.

Working: The angular speed of rotation of the toothed-wheel was increased from zero to a value ω until the light passing through one cut would completely be blocked by the adjacent tooth. This is ensured by the disappearance of the light while looking through the partially silvered glass plate.

Expression for speed of light: The speed v of light in air is equal to the ratio of the distance $2d$ (the distance light travelled from the toothed-wheel to the mirror and back), to the time taken t.

$$v = \frac{2d}{t} \qquad (6.12)$$

The distance d is a known value from the arrangement. The time t taken for the

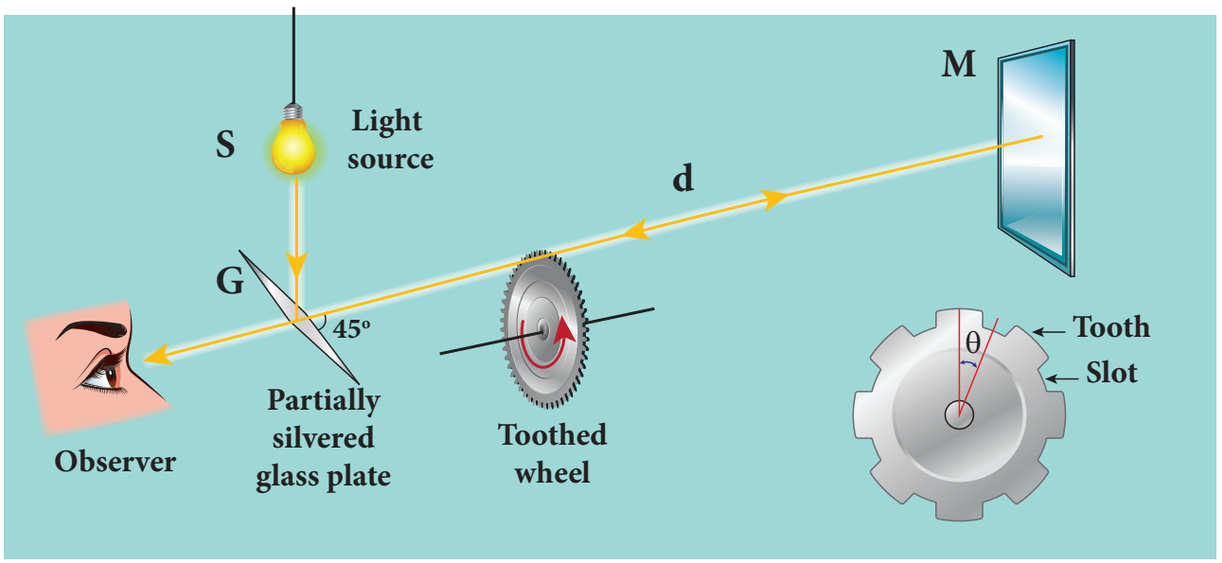


Figure 6.13 Speed of light by Fizeau's method

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